Status	Finished
Started	Wednesday, 21 October 2026, 4:08 PM
Completed	Wednesday, 21 October 2026, 4:12 PM
Duration	3 mins 45 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element
b. Checking each element sequentially ✓
c. Dividing the list in half repeatedly
d. Sorting the list before searching

Question **2**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O🤜
c. O(log n) ✓
d. O(1)

Question **3**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n ✓
b. n/2
c. 1
d. log n

Question **4**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 500
b. 10 ✓
c. 100
d. 1000

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

a. When the list is very small
b. When the list contains duplicate elements
c. When the list is unsorted
d. When the list is sorted ✓

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search continues from the beginning
b. The list is sorted
c. The search stops
d. The search continues in the left or right sublist ✓

Question **7**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and unsorted ✓
b. When the list is small and sorted
c. When the list is large and unsorted
d. When the list is large and sorted

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Bubble search
b. Merge search
c. Quick search
d. Linear search ✓

Question **9**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Complexity

b. Selection

c. Insertion

d. Bubble ✓

Question **10**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the middle of the list
b. In the left sublist
c. In the right sublist ✓
d. At the beginning of the list

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

b. Binary search algorithm is not efficient when the data elements more than 1500 ✓

c. There must be a mechanism to access middle element directly

d. Must use a sorted array

Question **12**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Hashing

b. Binary search

c. Jump search

d. Linear search ✓

Question **13**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(n log n)

b. O👎 ✓

c. O(log n)

d. O(1)

Question **14**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Deleting elements from a list

b. Sorting elements in a list

c. Determining whether an element is present in a list ✓

d. Inserting elements into a list

Question **15**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

a. Both (1) & (3)

b. Hash

c. Linear

d. Binary ✓